The exact final gradient curve of constant step gradient descent

Abstract

For a convex function with an L-Lipschitz gradient, we prove that the exact worst-case norm of the gradient after N constant steps of size h/L, under an initial-distance bound R, is $LR\max\{(1+Nh)^{-1}, |1-h|^N\}$ for $0 \le h \le 2$. This establishes the smooth-convex specialization of the final-gradient conjecture of Taylor, Hendrickx and Glineur. Existing objective and gradient bounds settle all even horizons and part of the odd-horizon range. A scaling argument transfers the known balanced history certificate to the progress branch. We close the remaining odd-horizon strips by continuing a published factorization and correcting at most two adjacent interpolation pairs with nonnegative squared norms. Every coefficient sign is proved uniformly in the horizon. The two classical scalar examples attain the bound. A direct Moreau-envelope corollary gives the corresponding exact relaxed-proximal residual curve.

Keywords: gradient descent; exact worst-case bounds; smooth convex interpolation; proximal residual

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1 The sharp bound

Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable and convex, with an L-Lipschitz gradient, where $L > 0$ and d is any positive integer. Assume that f has a minimizer x_*. We study the unchanged iteration

$$x_{i+1} = x_i - \frac{h}{L}\nabla f(x_i), \qquad i = 0, \dots, N-1. \tag{1}$$

The performance criterion is the gradient at x_N, after N updates, normalized by the initial distance to x_*.

Theorem 1. *For every integer $N \ge 1$, every $0 \le h \le 2$, and every trajectory (1),*

$$\|\nabla f(x_N)\| \le L\|x_0 - x_*\| \max\left\{\frac{1}{1+Nh}, |1-h|^N\right\}. \tag{2}$$

For each N, h, L and $R > 0$, equality in the worst-case bound with $\|x_0 - x_*\| \le R$ is attained by a one-dimensional function in the stated class.

The formula is the smooth-convex limit printed after Conjecture 3 of Taylor, Hendrickx and Glineur [1, Section 4.1.3]. Their interpolation framework and Huber-type and quadratic examples provide the original conjecture and the matching lower bounds. The theorem here does not address their strongly convex conjecture.

Recent work establishes the joint terminal certificate at the uniquely balanced constant step. Grimmer, Shu and Wang [2] introduced the associated composability framework and history inequality. Zhao [3] and Chikhi [4] proved the balanced constant-step statement in all finite horizons. Chikhi also extracted the sharp final-gradient optimum and its uniqueness, while leaving the whole curve away from the optimum open. We use and explicitly continue the A, D factorization of [4]; neither that factorization nor the balanced theorem is introduced here.

Two further predecessors are essential. Kim [5] proves the exact objective bound under initial distance, and the final-gradient bound under initial function gap is due to Rotaru, Glineur and Patrinos [6], with another proof in [5]. Their criteria differ from (2), but composing the two inequalities immediately settles every even horizon. Kim's curvature-comparison argument also provides methodological ancestry for transferring an optimized certificate to other parameters.

The remaining work is an odd-horizon interval between two consecutive balancing steps. Our contribution is an explicit correction of the continued factorization on that interval, together with an all-horizon proof that the corrected coefficients are nonnegative. The correction uses the established interpolation-plus-squares certificate method; its particular coefficients and feasibility proof are the ingredients needed here. We include the coefficient algebra and the finite exact arithmetic underlying one elementary root bound. No numerical performance-estimation output is used as a proof.

2 Interpolation and reductions

It suffices for the upper bound to work with $L = 1$, $x_* = 0$, and $f_* = 0$, by translating the optimizer, subtracting the minimum, and dividing the function by L. Write $f_i = f(x_i)$ and $g_i = \nabla f(x_i)$. Smooth convex interpolation gives

$$Q_{ij} := 2(f_i - f_j) - 2\langle g_j, x_i - x_j\rangle - \|g_i - g_j\|^2 \geq 0. \tag{3}$$

The optimizer index $*$ has $x_* = g_* = f_* = 0$. In particular,

$$\Phi_i := f_i - \tfrac{1}{2}\|g_i\|^2 \geq 0, \qquad \|g_i\| \leq \|x_i\|. \tag{4}$$

For $0 < h \leq 2$, gradient norms are nonincreasing. Indeed, with $d_i = g_{i+1} - g_i$, cocoercivity gives $\langle d_i, g_i\rangle \leq -\|d_i\|^2/h$, hence

$$\|g_{i+1}\|^2 - \|g_i\|^2 \leq -\frac{2-h}{h}\|d_i\|^2 \leq 0. \tag{5}$$

For $m \geq 1$, let $b_m = 1 + \rho_m \in (1, 2)$ be the unique solution of

$$\rho_m^m(1 + mb_m) = 1. \tag{6}$$

These roots increase strictly with m; also $b_1 = \sqrt{2}$ and $b_2 = 3/2$. To see the ordering, at a root $r = \rho_m$ the strictly concave function $u \mapsto u \log r + \log(1 + u(1 + r))$ vanishes at 0 and m, and is negative at $m + 1$. Thus $\rho_{m+1} > \rho_m$.

Lemma 1 (Progress branch)**.** *For every $N \geq 1$ and $0 < h \leq b_N$, put $S = Nh$. Then*

$$\|x_N\|^2 + 2Sf_N + S(S+1)\|g_N\|^2 \leq \|x_0\|^2, \qquad \|g_N\| \leq \frac{\|x_0\|}{1+Nh}. \tag{7}$$

Proof. The balanced history inequality of [3, 4], also re-established by the coefficient calculation below, states that

$$\mathcal{H}_b(F) := 2b\sum_{i<N}(F_i - F_N) - b(b-1)\sum_{i<N}\|G_i\|^2 - Nb(1+Nb)\|G_N\|^2 \geq 0 \tag{8}$$

where $b = b_N$, $F_i = F(y_i)$, $G_i = \nabla F(y_i)$, and $y_{i+1} = y_i - bG_i$ for a convex 1-smooth F. Apply it to $F = (h/b)f$. This is 1-smooth and has exactly the original trajectory. If $\mathcal{H}_h(f)$ denotes the same expression with b replaced by h, direct substitution yields

$$\mathcal{H}_h(f) - \mathcal{H}_b((h/b)f) = h(1-h/b)\sum_{i<N}\big(\|g_i\|^2 - \|g_N\|^2\big) \geq 0$$

by (5). Expanding the recurrence gives the exact identity

$$\|x_0\|^2 - \|x_N\|^2 - 2Sf_N - S(S+1)\|g_N\|^2 = \mathcal{H}_h(f) + h\sum_{i<N} Q_{*i}. \tag{9}$$

This proves the energy. Its left-hand energy is at least $(1+S)^2\|g_N\|^2$ by (4). ∎

The next two inequalities are the precise external bounds used in the proof. For $m \geq 1$ and $0 < h < 2$, set

$$M_m(h) = \max\{(1+2mh)^{-1}, |1-h|^{2m}\}, \qquad M_0(h) = 1. \tag{10}$$

Kim's objective theorem and the Rotaru–Glineur–Patrinos gradient theorem, in the normalization above, give

$$f(x_m) - f_* \leq \tfrac{1}{2}M_m(h)\|x_0 - x_*\|^2, \quad \|g_m\|^2 \leq 2M_m(h)(f(x_0) - f_*). \tag{11}$$

See [5, Theorems 1.3 and 1.5] and [6, Theorem 2.2], specialized to zero strong convexity. For $m = 0$ these inequalities are standard smoothness inequalities. Applying the first to a prefix of length a and the second to the actual remaining trajectory of length b gives

$$\|g_N\|^2 \leq M_a(h)M_b(h)\|x_0\|^2, \qquad a + b = N. \tag{12}$$

If $N = 2m$, the choice $a = b = m$ is exactly (2). If $N = 2m+1$ and $h \geq b_{N+1}$, both factors with $a = m, b = m+1$ are on their oscillatory branches, so (12) gives $\|g_N\| \leq (h-1)^N\|x_0\|$. Thus only

$$N \text{ odd}, \qquad b_N \leq h \leq b_{N+1} \tag{13}$$

requires a new certificate. The overlap at the endpoints is intentional.

3 A continued interpolation certificate

We first work for any integer $N \geq 1$ and $b_N \leq h < 2$, without assuming that N is odd. Put

$$r = h-1, \quad S = Nh, \quad q = r^{-N}, \quad \Delta = S+1-q \geq 0, \quad t = q-1, \quad p_k = \frac{r^{-2k}-1}{2}. \tag{14}$$

The following sequences and pair weights are the factorization in [4, Section 4], evaluated away from its balancing root:

$$A_{-1} = 0, \quad A_0 = 1, \quad D_1 = p_1, \quad D_{N+1} = 0, \tag{15}$$

$$\frac{A_k}{A_{k-1}} = \frac{(1+kh+r^{-k})(1+r^{-(k+1)})}{(1+r^{-k})(1+kh-r^{-k})}, \tag{16}$$

$$\frac{D_{k+1}}{D_k} = \frac{(1+kh-r^{-k})(r^{-(k+1)}-1)}{(r^{-k}-1)(1+kh+r^{-k})}, \qquad 1 \leq k < N. \tag{17}$$

For $N = 1$ the recurrences are empty. For $0 \leq a < b \leq N$, define

$$\lambda_{ab} = (A_a - A_{a-1})(D_b + rD_{b+1}), \qquad \lambda_{ba} = (A_{a-1} + rA_a)(D_b - D_{b+1}). \tag{18}$$

Finally set

$$T = \frac{(q+1)\Delta}{2}, \qquad c = \frac{(q-1)\Delta}{2}, \qquad a_i = T\frac{A_i - A_{i-1}}{A_{N-1}} \quad (0 \leq i < N). \tag{19}$$

Thus $T - c = \Delta$.

Lemma 2. *Every denominator above is positive. The A_k strictly increase, the D_k strictly decrease through $D_N > 0$, and all λ_{ij} and a_i are nonnegative. With*

$$\mathcal{E}_t(x_N) = \|x_N\|^2 + 2tf_N + t(t+1)\|g_N\|^2,$$

the following identity holds under the recurrence:

$$\|x_0\|^2 - \mathcal{E}_t(x_N) = \sum_{i \neq j} \lambda_{ij} Q_{ij} + h \sum_{i<N} Q_{*i} + 2\sum_{i<N} a_i \Phi_i - 2c\Phi_N. \tag{20}$$

The pair sum is over $0 \leq i, j \leq N$.

Proof. The function $(r^{-u}-1)/u$ strictly increases for $u > 0$. Since $q \leq 1 + Nh$, this gives $1 + kh > r^{-k}$ for $1 \leq k < N$. For $s = 1 + kh$, $u = r^{-k}$ and $\delta = 1 - r$, subtraction in the recurrence ratios gives

$$\frac{A_k}{A_{k-1}} - 1 = \frac{u[\delta s + hu + 2r]}{r(s-u)(1+u)} > 0,$$

$$1 - \frac{D_{k+1}}{D_k} = \frac{hu[u - 1 - k\delta]}{r(u-1)(s+u)} > 0.$$

The second sign follows from $u = (1+\delta/r)^k \geq 1 + k\delta/r > 1 + k\delta$. Multiplication of the ratios gives

$$A_{k-1} D_k = p_k, \qquad 1 \leq k \leq N. \tag{21}$$

These facts prove all initial sign assertions.

For completeness, the two sum identities needed off balance are

$$\frac{\sum_{i<k} A_i}{A_k} = \frac{1 + kh - r^{-k}}{h(1 + r^{-(k+1)})}, \tag{22}$$

$$\frac{\sum_{j=k+2}^{N} D_j}{D_{k+1}} = \frac{1 + (k+1)h - r^{-(k+1)}}{h(r^{-(k+1)} - 1)} - \frac{D_N}{D_{k+1}} \frac{\Delta}{h(q-1)}, \tag{23}$$

for $0 \leq k < N$. The first follows by forward induction, beginning at the empty sum $k = 0$. The second follows by backward induction, beginning at $k = N - 1$, where its two terms cancel. Substituting (16)–(17) verifies each induction step. The last term in (23) is the boundary correction that vanishes at balance.

Let $d_i = \sum_{j \neq i} (\lambda_{ij} - \lambda_{ji})$. Summing across the cut after k and using (21)–(23) yields

$$\begin{aligned} \sum_{i=0}^{k} d_i &= (1-r) p_{k+1} + h \left(A_k \sum_{j=k+2}^{N} D_j - D_{k+1} \sum_{i<k} A_i \right) \\ &= h(k+1) - \frac{D_N \Delta}{q-1} A_k. \end{aligned}$$

Because $A_{N-1} D_N = p_N$, it follows that

$$d_i = h - a_i \ (i < N), \qquad d_N = -S + T. \tag{24}$$

Every mixed gradient coefficient cancels. Indeed, for $a < b$,

$$\begin{aligned} \lambda_{ab} + \lambda_{ba} &= h(A_a D_b - A_{a-1} D_{b+1}), \\ \sum_{i \leq a} \lambda_{ib} &= A_a (D_b + r D_{b+1}), \qquad \sum_{i > b} \lambda_{ia} = (A_{a-1} + r A_a) D_{b+1}. \end{aligned}$$

After inserting $x_i = x_0 - h \sum_{j<i} g_j$ into (3), half the coefficient of $\langle g_a, g_b \rangle$ is

$$\lambda_{ab} + \lambda_{ba} - h \sum_{i \leq a} \lambda_{ib} + h \sum_{i > b} \lambda_{ia} = 0.$$

For the diagonal coefficients, put $U_k = \sum_{i<k} \lambda_{ik}$ and $V_k = \sum_{i>k} \lambda_{ik}$. Telescoping and (21) give

$$r V_k - U_k = r^2 p_{k+1} - p_k = \frac{1 - r^2}{2} \qquad (k < N).$$

Thus the coefficient of $\|g_k\|^2$ is $-d_k - 2U_k + 2rV_k = -hr + a_k$. The terminal coefficient is $-d_N - 2p_N = S + 1 - T - q^2$. Consequently

$$\begin{aligned} \sum_{i \neq j} \lambda_{ij} Q_{ij} &= 2 \sum_{i<N} (h - a_i) f_i - 2(S - T) f_N \\ &\quad + \sum_{i<N} (a_i - hr) \|g_i\|^2 + (S + 1 - T - q^2) \|g_N\|^2. \end{aligned}$$

Adding $h\sum Q_{*i} + 2\sum a_i\Phi_i$ and expanding the recurrence leaves

$$\|x_0\|^2 - \|x_N\|^2 - 2(S-T)f_N - (q^2 - 1 - S + T)\|g_N\|^2.$$

Finally $S - T + c = t$ and $q^2 - 1 - S + T - c = t(t+1)$, proving (20). ∎

At $h = b_N$, one has $\Delta = 0$, $q = 1 + S$ and $a_i = c = 0$. The pairwise coefficient calculation in the proof is then exactly $\sum_{i\neq j}\lambda_{ij}Q_{ij} = \mathcal{H}_{b_N}(f)$. This proves (8) with the credited balanced factorization, so the use of the history inequality in Lemma 1 is also self-contained.

4 The adjacent square correction

Return to the closed odd strip (13). Define

$$C_i = \sum_{j=0}^{i} a_j = T\frac{A_i}{A_{N-1}}, \qquad J_i = [C_i - \Delta]_+, \qquad J_{-1} = 0, \tag{25}$$

$$\beta_i = a_i - J_i + J_{i-1}, \qquad \kappa_i = \frac{2J_i}{h} \qquad (0 \le i < N). \tag{26}$$

Here $[u]_+ = \max\{u, 0\}$. Since $T - \Delta = c$, one has

$$J_{N-1} = c, \qquad \beta_i = \min\{C_i, \Delta\} - \min\{C_{i-1}, \Delta\} \ge 0, \tag{27}$$

with $C_{-1} = 0$. Modify only adjacent pair weights:

$$\widehat{\lambda}_{i,i+1} = \lambda_{i,i+1} + \frac{J_i}{h}, \qquad \widehat{\lambda}_{i+1,i} = \lambda_{i+1,i} - \frac{rJ_i}{h}. \tag{28}$$

For nonadjacent pairs put $\widehat{\lambda}_{ij} = \lambda_{ij}$.

Lemma 3. *The exact identity*

$$\begin{aligned}\|x_0\|^2 - \mathcal{E}_t(x_N) = &\sum_{i\neq j}\widehat{\lambda}_{ij}Q_{ij} + h\sum_{i<N}Q_{*i} + 2\sum_{i<N}\beta_i\Phi_i \\ &+ \sum_{i<N}\kappa_i\|g_{i+1} + rg_i\|^2\end{aligned} \tag{29}$$

holds for the recurrence.

Proof. A direct expansion of one step gives

$$Q_{i,i+1} - rQ_{i+1,i} + 2\|g_{i+1} + rg_i\|^2 = 2h(\Phi_i - \Phi_{i+1}). \tag{30}$$

The function-value terms on the left are $2h(f_i - f_{i+1})$; its diagonal gradient terms are $-h\|g_i\|^2 + h\|g_{i+1}\|^2$, and its mixed term vanishes. Multiply (30) by J_i/h and sum. Replacing a_i by β_i cancels every interior Φ_i change, leaving $-2J_{N-1}\Phi_N = -2c\Phi_N$. Equation (20) now proves (29). ∎

The only weights whose signs are not immediate are the reverse adjacent weights in (28). Their feasibility is precisely

$$J_i \le \frac{h}{r}\lambda_{i+1,i}, \qquad 0 \le i < N. \tag{31}$$

We next prove this uniformly in the horizon.

5 Feasibility on every odd strip

We use the following elementary estimate, with its exact finite base cases and analytic tail proved in the Appendix.

Lemma 4. *For every integer $m \ge 1$ and $\rho_m \le r < 1$,*

$$m(1-r)^2 \le \frac{2}{3}. \tag{32}$$

Throughout this section write $\delta = 1 - r$ and $s_k = 1 + kh$. The strip inequalities and a useful upper bound on r are

$$r(s_N + h) \le q \le s_N, \qquad r \le \frac{N}{N+1}. \tag{33}$$

The first two inequalities follow directly from (6) and the endpoints of the strip. For the last, set $m = N + 1 \ge 2$. The function $(1 - 1/m)^m$ increases for real $m \ge 2$, by differentiation, and its value at 2 is $1/4$. Hence

$$(1-1/m)^m\big(1 + m(2 - 1/m)\big) = 2m(1-1/m)^m \ge m/2 \ge 1.$$

The increasing root polynomial gives $\rho_{N+1} \le N/(N+1)$.

Lemma 5. *If $N \ge 3$ is odd, then $J_i = 0$ for $i \le N-3$. Thus at most two squares in* (29) *have a nonzero coefficient.*

Proof. If $\Delta = 0$, every flow vanishes. Otherwise, since A_i increases, it is enough to show

$$(q+1)\frac{A_{N-3}}{A_{N-1}} \le 2.$$

Put $s = s_{N-1}$, $u = rq$, and $v = r^2 q$. Multiplying the last two ratios in (16) gives

$$(q+1)\frac{A_{N-3}}{A_{N-1}} = (1+v)\frac{s-h-v}{s-h+v}\frac{s-u}{s+u}. \tag{34}$$

All factors are positive. Equation (33) implies $u \ge r^2(s_N + h)$ and $v \ge r^3(s_N + h)$, so the last two ratios are bounded by $(1-r^3)/(1+r^3)$ and $(1-r^2)/(1+r^2)$, respectively. Also

$$1 + v \le 1 + r^2 + Nhr^2 \le N(1+r^2). \tag{35}$$

For the second inequality it suffices that $N(1-r^3) \geq 1+r^2$. By (33),

$$N(1-r^3) \geq \frac{N}{N+1}(1+r+r^2) \geq 1+r^2.$$

The last step is equivalent to $Nr \geq 1+r^2$, which holds for $N \geq 3$ and $1/2 \leq r < 1$; here $r \geq \rho_3 > \rho_2 = 1/2$. Combining these bounds in (34) gives

$$(q+1)\frac{A_{N-3}}{A_{N-1}} \leq N(1-r)^2\frac{1+r+r^2}{1-r+r^2} \leq 3N(1-r)^2 \leq 2.$$

The middle inequality is equivalent to $2(1-r)^2 \geq 0$, and the last is Lemma 4. ∎

Lemma 6. *All capacities* (31) *hold on every closed odd strip, including* $N = 1$.

Proof. For the final edge, including $N = 1$ with $A_{-1} = 0$,

$$\lambda_{N,N-1} = (A_{N-2} + rA_{N-1})D_N \geq rp_N.$$

It therefore suffices that $\Delta \leq h(q+1)$, since $J_{N-1} = (q-1)\Delta/2$. From (33),

$$\Delta \leq \delta s_N - rh \leq \delta s_N, \qquad q \geq rs_N.$$

Moreover $r \geq \sqrt{2}-1$ implies $\delta \leq hr$. Thus $\Delta \leq hrs_N \leq hq < h(q+1)$. This settles $N = 1$ completely.

Suppose now $N \geq 3$. Lemma 5 leaves only the penultimate edge. Put

$$m = N-1, \qquad s = 1+mh, \qquad u = r^{-m} = rq, \qquad e = s-u > 0.$$

The last ratio in (16) gives exactly

$$J_{N-2} = \Delta\left[\frac{s(u-1)-u(u+3)}{2(s+u)}\right]_+. \tag{36}$$

If $N = 3$, then $s = 3+2r < 5$ and $u > 1$. The numerator is bounded above by

$$5(u-1)-u(u+3) = -(u-1)^2 - 4 < 0.$$

Thus $J_1 = 0$ and no penultimate correction is needed in this case.

For $N \geq 5$, rewrite the numerator in (36) as $(u-1)e - 4u$ to obtain

$$J_{N-2} \leq \frac{\Delta(u-1)e}{2(s+u)}. \tag{37}$$

By (21) and the D ratio,

$$\begin{aligned}\frac{h}{r}\lambda_{N-1,N-2} &\geq hp_m\left(1-\frac{D_N}{D_{N-1}}\right)\\ &= \frac{h^2u(u+1)(u-1-m\delta)}{2r(s+u)} \geq \frac{h^2\delta u(u+1)(u-1)}{2r(s+u)}.\end{aligned} \tag{38}$$

The final inequality follows from Bernoulli: $u \geq 1 + m\delta/r$ implies $u - 1 - m\delta \geq \delta(u-1)$. Next (33) yields

$$\Delta \leq \delta s_N, \qquad e = s_N - h - rq \leq h\delta s_N - h(1+r^2) \leq h\delta s_N, \qquad u \geq r^2 s_N.$$

Consequently

$$\Delta e \leq h\delta^2 s_N^2, \qquad \frac{h^2\delta}{r} u(u+1) \geq h^2 \delta r^3 s_N^2. \tag{39}$$

Finally $\rho_5 > 5/8$, since

$$(5/8)^5\big(1 + 5(1+5/8)\big) = \frac{228125}{262144} < 1.$$

The function $(1+r)r^3 - (1-r)$ strictly increases for $r > 0$ and equals $89/4096 > 0$ at $5/8$. Hence $r \geq \rho_N \geq \rho_5$ gives $\delta \leq hr^3$. Equations (39) now show $\Delta e \leq (h^2\delta/r)u(u+1)$. Comparing (37) and (38) proves the remaining capacity. Every inequality permits both strip endpoints, and none divides by Δ. ∎

Theorem 2 (Odd-strip energy)**.** *For every odd $N \geq 1$ and $b_N \leq h \leq b_{N+1}$, the normalized trajectory satisfies*

$$\|x_N\|^2 + 2(q-1)f_N + q(q-1)\|g_N\|^2 \leq \|x_0\|^2, \qquad q = (h-1)^{-N}. \tag{40}$$

In particular $\|g_N\| \leq (h-1)^N \|x_0\|$.

Proof. Lemmas 2 and 6, together with (27), make every coefficient on the right of (29) nonnegative. All interpolation terms, Φ_i, and squared norms are nonnegative. This proves the energy. By (4), the energy is at least $[1 + (q-1) + q(q-1)]\|g_N\|^2 = q^2\|g_N\|^2$. At $h = b_N$, all repair coefficients vanish. For $N = 1$ all interior recurrences and early-support assertions are empty, and only the already proved final-edge capacity is used. ∎

6 Assembly, equality, and the proximal consequence

Proof of Theorem 1. For $0 < h < 2$, every even horizon follows from (12) with equal block lengths. For odd N, Lemma 1 covers $h \leq b_N$, Theorem 2 covers $b_N \leq h \leq b_{N+1}$, and (12) covers $h \geq b_{N+1}$. The same split is valid for $N = 1$ using $M_0 = 1$.

For $0 \leq h \leq 1$, $(1-h)^N(1+Nh) \leq 1$, by $(1-h)^N \leq (1+h)^{-N} \leq (1+Nh)^{-1}$. For $1 \leq h \leq b_N$, the root equation and monotonicity give $(h-1)^N(1+Nh) \leq 1$; the inequality reverses for $h \geq b_N$. Thus the upper bounds in the three regions are exactly the maximum in (2). At $h = 0$, the bound is $\|\nabla f(x_0)\| \leq L\|x_0 - x_*\|$. At $h = 2$, equation (5) gives the same bound for g_N. These arguments check both endpoints directly.

Restore arbitrary L by applying the normalized result to f/L, and restore x_* by translation. No dimension-dependent argument was used. For sharpness, fix $R > 0$ and

put $a = R/(1+Nh)$. The scalar Huber function

$$f(x) = \begin{cases} Lx^2/2, & |x| \le a, \\ L(a|x| - a^2/2), & |x| \ge a \end{cases} \tag{41}$$

is differentiable, convex and L-smooth. Starting at $x_0 = R$ gives $x_i = R - iha \ge a$ for $i \le N$, and $x_N = a$; hence $|\nabla f(x_N)| = LR/(1+Nh)$. The scalar quadratic $f(x) = Lx^2/2$ gives $x_N = (1-h)^N R$ and $|\nabla f(x_N)| = LR|1-h|^N$. Selecting the larger example proves exactness, including $h = 0, 2$. Either example embeds into every positive dimension by depending only on one coordinate. ∎

The following is a direct Moreau-envelope transfer, already used for the neighboring and balanced results in [4, 7]. It is not a separate certificate construction.

Corollary 6.1. *Let $F : \mathbb{R}^d \to \mathbb{R} \cup \{+\infty\}$ be proper, closed and convex, with a minimizer x_*. Fix $N \ge 1$ and $\lambda > 0$, and run relaxed proximal point with a constant $0 \le h \le 2$:*

$$z_i = \mathrm{prox}_{\lambda F}(x_i), \qquad x_{i+1} = x_i + h(z_i - x_i), \qquad 0 \le i < N.$$

With $z_N = \mathrm{prox}_{\lambda F}(x_N)$,

$$\left\| \frac{x_N - z_N}{\lambda} \right\| \le \frac{\|x_0 - x_*\|}{\lambda} \max\left\{ \frac{1}{1+Nh}, |1-h|^N \right\}. \tag{42}$$

The normalized worst-case residual is exactly the displayed maximum.

Proof. The Moreau envelope $F_\lambda(x) = \inf_z\{F(z) + \|z-x\|^2/(2\lambda)\}$ is convex and differentiable, has a $1/\lambda$-Lipschitz gradient, and has the same minimizers as F. Its gradient is $\nabla F_\lambda(x) = (x - \mathrm{prox}_{\lambda F}(x))/\lambda$. Thus the stated recurrence is exactly gradient descent on F_λ with normalized step h and smoothness constant $1/\lambda$. Theorem 1 gives (42).

For the progress lower example take $x_0 = R > 0$, $x_* = 0$ and $F(x) = a|x|$, where $a = R/[\lambda(1+Nh)]$. Soft thresholding gives $x_i = R - ih\lambda a \ge \lambda a$, $x_N = \lambda a$ and $z_N = 0$. For the oscillatory example take F to be the indicator of $\{0\}$. Then $z_i = 0$ and $x_N = (1-h)^N R$. These examples have residuals equal to the two branches in (42). ∎

A An exact uniform bound for the balancing roots

Proof of Lemma 4. It suffices to work at $r = \rho_m$. For $1 \le m \le 23$, take $D = 1000$ and the following fixed integers d_m:

m	1	2	3	4	5	6	7	8	9	10	11	12
d_m	816	577	471	408	365	333	308	288	272	258	246	235

m	13	14	15	16	17	18	19	20	21	22	23
d_m	226	218	210	204	198	192	187	182	178	174	170

These finite rational certificates satisfy the two exact integer inequalities

$$3md_m^2 \le 2D^2, \qquad (D-d_m)^m[D+m(2D-d_m)] < D^{m+1}. \tag{43}$$

Every entry is specified, so (43) is verified by integer powers and comparisons, without evaluating a balancing root. The second inequality says that $1-d_m/D < \rho_m$. The first then gives $m(1-\rho_m)^2 \le 2/3$.

For all $m \ge 24$, put $x = \sqrt{2m/3} \ge 4$ and $\varepsilon = \sqrt{2/(3m)}$, so $m\varepsilon = x$. At $r_0 = 1-\varepsilon$,

$$r_0^m[1+m(1+r_0)] \le e^{-x}(1+3x^2-x) < e^{-x}(1+3x^2) < 1.$$

For the final strict inequality,

$$e^4 > \sum_{k=0}^{7} \frac{4^k}{k!} = \frac{16319}{315} > 49,$$

and $e^x/(1+3x^2)$ strictly increases for $x \ge 4$, because its logarithmic derivative is $1 - 6x/(1+3x^2) > 0$. Therefore $\rho_m > 1-\varepsilon$ and $m(1-\rho_m)^2 < 2/3$ for every $m \ge 24$. This analytic tail and the specified finite certificates cover every positive integer. For $\rho_m \le r < 1$, the quantity $(1-r)^2$ can only decrease. ∎

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